Noel Burton-Krahn

Scientific Software Engineer / Consultant

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Remote - Canada, EU

PROFILE

Scientific software engineer with more than 30 years of experience building algorithms and reliable software for biomedical devices, scientific instruments, autonomous systems, and distributed infrastructure. Hands-on C/C++ and Python expertise spans ECG/EEG signal processing, machine learning, Linux, and real-time systems, from research prototypes through regulated products.

TECHNICAL FOCUS

Scientific and biomedical: algorithms, signal processing, ECG/EEG, physiological data, machine learning/deep learning (PyTorch, U-Net), numerical and research software

Robotics and instrumentation: PX4 autonomy, real-time control, hardware integration, radio-astronomy correlator simulation, remote sensing

Systems: Linux, distributed systems, fault tolerance, networking, OpenStack/SDN, cloud infrastructure, data systems

Languages and tools: C/C++, Python, Matlab/Octave, Perl, C#/.NET, Java, SQL, PyTorch, Git, Make/CMake, Docker, Kubernetes, Google Cloud, AWS, Cursor

RECENT SCIENTIFIC & TECHNICAL EXPERIENCE

ECG Algorithm Development Contractor · HeartEye — May 2026 – Oct 2026

Developed a PQRST detection algorithm in Python and C++ for a 12-lead embedded ECG device, combining conventional signal processing with a PyTorch U-Net model.

Software Team Lead · Avy Drone Networks — Nov 2024 – Aug 2025

Led and mentored the software team developing an autonomous fixed-wing aircraft, coordinating software projects and working closely with hardware engineers. Supervised four developers and hired two more.

Contributed as a senior developer to the PX4-based system using C++ and Qt in a real-time, Linux-based robotics environment.

Head of Software · Prolira — May 2017 – Aug 2024

Lead software architect and developer; managed engineering, quality and regulatory processes, and technical documentation. Developed software for a medical device that received FDA approval.

Built a substantial EEG analysis and AI platform in C++ and Python, combining digital signal processing and machine learning.

Developed algorithms in MATLAB and ported MATLAB code to production C++.

Built Google Cloud data and ML-training infrastructure and a portal for remote device acquisition and customer support.

Data Scientist · Clinezeman — Apr 2017 – Jan 2019

Developed a machine-learning application for biomedical signal processing using C++, Python, Matlab/Octave, mathematical algorithms, and neural-network methods; this work continued into Prolira.

Senior Distributed Systems Engineer · Cisco / Piston — Jun 2013 – Apr 2017

Developed Linux and Python system-control software for OpenStack clusters, software-defined networking, and virtualization.

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SELECTED SCIENTIFIC AND SYSTEMS WORK

Lead Engineer · Gatekeeper Systems — Apr 2003 – Feb 2006

Delivered a portable Linux digital-video surveillance system and Windows playback software using C/C++, Perl, OpenGL, VB.NET, and SQLite.

Research Developer · CardioComm / Harley Street Software — May 1994 – Jun 1998

Developed Windows ECG applications and signal-processing software, including the P-QRS-T feature-detection algorithm covered by US Patent 5,758,654; C/C++, Visual Basic, MFC, OLE, and ActiveX.

SELECTED ALGORITHM AND SYSTEMS PROJECTS

Research Programmer · Dominion Radio Astrophysical Observatory — Jan 1992 – May 1992

Designed and developed a Matlab and C simulation of a hardware correlator combining radio-antenna signals on Sun SPARC and IBM RISC 6000 workstations.

Selected Independent Projects

HotSwap · Transparent Linux Server Failover — Published 2002

Developed a Linux kernel/server-replication system for seamless failover, preserving active TCP connections, internal memory state, and external network state. Presented at USENIX LISA XVI and covered by a US patent application.

BKBox and NatNix · Linux and Networking Products

Built a Linux office server providing OpenAFS remote files, secure IMAP, web services, and firewall/gateway functions; developed a C++, firewall-tunnelling peer-to-peer VPN for Linux and Windows.

EARLIER EXPERIENCE

2012–2013: Kal Tire — point-of-sale and inventory systems; C#/.NET, Groovy, SQL.

2008–2012: Clover Point, Telocate, Eightfold Logic — GIS, remote-device tracking, and analytics; Python, C#, Oracle, PostgreSQL, Perl.

2006–2008: Washington State DOC and Kodak — Java/WebSphere/DB2 mainframe integration and commercial-printing software.

1998–2003: Kronofusion, Mercurial, FaxPC, BC Government — encrypted TCP tunnelling, remote desktop, Linux/Windows applications, and technical leadership.

Co-op Software Engineer / Research Programmer · Power Measurement / Siemens; Pacific Forestry Centre — Sep 1992 – Sep 1993

Developed real-time C++ power-meter communications and a Prolog expert system integrating remote sensing, image processing, GIS, and modelling.

SELECTED PUBLICATIONS AND PATENTS

ECG P-QRS-T Onset, Offset and Peak Detection Method and Apparatus. US Patent 5,758,654, Harley Street Software, 1998.

HotSwap — Transparent Failover for Linux Servers. USENIX LISA XVI, 2002.

Method and apparatus for providing transparent fault tolerance within an application server environment. US Patent Application 2004/0153709, 2004.

EDUCATION

M.Sc., Computer Science, University of Victoria, 2003.
B.Sc. (Honours), Computer Science / Mathematics / Physics, University of Victoria, 1994

REFERENCES

Available on request